

Demand Approximation Framework

Factor Reference, Dynamic Weighting & Human Judgment Guide

v2.0 · Pricing Analytics · Draft · Dual-use: Human & AI System

PURPOSE OF THIS DOCUMENT This document is structured for two audiences simultaneously: 1. HUMANS (pricing analysts, product managers, leadership) — use Sections 1–3 to understand what the model does, why each factor is included, and where human judgment must override it. 2. AI SYSTEMS & ENGINEERS — every factor includes a machine-readable formula, data schema, reliability score, and fallback rule. Section 4 defines the complete end-to-end computation. Section 5 lists factors that must never be modelled automatically.

13	7	4	2
Total factors	In model	Human-judged	Unquantifiable
across all tiers	quantifiable, R ² -weighted	flag before deciding	never modelled

1. Master demand projection formula

Every what-if projection resolves to one expression. All factors below multiply into it. This is the single formula an AI system should execute.

$$Q_{new} = Q_{base} \times (P_{new} / P_{base})^{e_{eff}} \times S \times I \times (1 + lift_M)$$

$$e_{eff} = e \times C \times L \times X$$

$$Profit = (P_{new} - Cost) \times Q_{new}$$

Sym bol	Factor	Type	Default if unavailable
e	Price elasticity	Core exponent — shapes the entire curve	Must estimate — no fallback
S	Seasonality	Multiplicative — shifts baseline up/down	1.0 (no seasonal adjustment)
C	Competitor pricing	Elasticity modifier — makes e more/less steep	1.0 (neutral)
M	Promotions	Additive uplift on projected Q	0 (no promotion active)
I	Inventory signal	Multiplicative — scarcity or censoring fix	1.0 (normal stock)
L	Product lifecycle	Elasticity modifier — adjusts for product age	1.0 (mature product)
X	Consumer sentiment	Elasticity modifier — macro demand climate	1.0 (neutral CCI)

Dynamic weight formula — applied to all factors

$$w_i = R2_i / (R2_e + R2_S + R2_C + R2_M + R2_I + R2_L + R2_X)$$

AI SYSTEM RULE: If a factor's R2 = 0 or data is missing, set factor value = 1.0 and exclude it from the denominator. Renormalise remaining weights. Never set a factor to 0 — multiplying by 0 collapses the entire projection. Re-run weight normalisation every 4-week data batch.

Complete worked example

Inputs: P_base = 200, Q_base = 100, Cost = 80, P_new = 220 (+10%) e = -1.5 | S = 1.2 (mild seasonal peak) | C: gap = +0.16 -> modifier = 1.032 M = 0 (no promo) | I = 1.0 (normal stock) | L = 0.85 (growth phase) | X = 1.0 e_eff = -1.5 x 1.032 x 0.85 x 1.0 = -1.316 Q_new = 100 x (220/200)^(-1.316) x 1.2 x 1.0 = 100 x 0.877 x 1.2 = 105.2 units Profit = (220 - 80) x 105.2 = 14,728 vs base profit (200-80)x100 = 12,000 -> +22.7%

2. The 7 modelled factors

Each factor below is quantifiable from data. Its contribution is automatically weighted by its R2 from regression. Factors with stronger explanatory power receive higher weight.

F1 — Price elasticity (e)

Criticality: CRITICAL — Cannot be omitted. All other factors modify or multiply into e.

Dynamic weight formula

$$w_e = R2_OLS / \text{sum}(R2_all)$$

Start $w_e = 1.0$. After first regression, replace with $R2_OLS$. Recalculate every 4-week batch.

How to find the value

Model	Log-linear OLS: regress $\log(Q)$ on $\log(P)$. Slope = e. Use Double ML when promos or seasonality confound.
Minimum data	30 clean price-quantity observations. Weekly granularity preferred.
Rolling update	Sliding 8-12 week window. Flag if $ e $ shifts > 0.5 between windows.
Range check	Normal: -0.2 to -4.0. Flag and investigate if $ e > 5$.
AI rule	If fewer than 10 obs, use category proxy e. Label estimate as PROVISIONAL.

Data source

- Internal: weekly sales extract — date, sku_id, unit_price, units_sold.
- Kaggle: Retail Price Optimization (suddharshan), Avocado Prices (neuromusic).

Example

Base: $P=200$, $Q=100$. $e = -1.5$. Raise price 10% -> $Q_new = 100 \times (220/200)^{-1.5} = 85.7$ units. Revenue: $220 \times 85.7 = 18,854$ vs $200 \times 100 = 20,000$ -> -5.7%. Note: revenue falls but margin per unit rises. Profit impact depends on cost.

F2 — Seasonality (S)

Criticality: HIGH — Can shift demand +/-50%. Ignoring it inflates or deflates e by up to 40%.

Dynamic weight formula

$$S_t = \text{avg_demand_period_t} / \text{overall_avg_demand}$$

$$w_S = R2_seasonal / \text{sum}(R2_all)$$

If $R2 < 0.05$, product is non-seasonal. Set $S = 1.0$ and exclude from weights.

How to find the value

Method	STL decomposition (statsmodels) or Fourier seasonal regression.
Min data	2 full seasonal cycles (2 years weekly). Use category index if history is thin.
AI rule	$S_t > 1$ = above-average period. $S_t < 1$ = below. Clamp to range 0.3 - 3.0.

Data source

- Internal: same weekly sales extract, aggregated by week-of-year across years.
- Kaggle: Store Sales Time Series Forecasting (Rossmann dataset).

Example

December seasonal index $S = 1.7$. Base projection = 100 units. Adjusted = $100 \times 1.7 = 170$ units. A 10% price rise in December: $85.7 \times 1.7 = 145.7$ — still above off-season baseline of 85.7.

F3 — Competitor pricing (C)

Criticality: MEDIUM — Adjusts e by +/-15-20%. Crucial when a competitor makes a large price move.

Dynamic weight formula

$$\text{Gap} = (\text{P}_{\text{own}} - \text{P}_{\text{comp_avg}}) / \text{P}_{\text{comp_avg}}$$

$$\text{C} = 1 + 0.2 \times \text{sign}(\text{Gap}) \times \min(|\text{Gap}|, 0.5)$$

$$\text{e}_{\text{eff}} = \text{e} \times \text{C}$$

If competitor data unavailable, set $\text{C} = 1.0$. Do not impute. Missing data = neutral assumption.

How to find the value

Scraping	Selenium/Scrapy on competitor pages, or price intelligence API (Prisync, Wiser).
Frequency	Weekly for FMCG/electronics. Monthly for slow-moving categories.
Regression	Include Gap as covariate in demand model. Its R2 contribution = w_{C} .
AI rule	Clamp Gap at +/-0.5 (50% price difference). Beyond that, relationship is non-linear.

Data source

- Internal: competitor price log if maintained.
- External: Prisync API, DataImpact, or manual tracking.
- Kaggle: Retail Price Optimization (bhanupratapbiswas) — includes competitor price columns.

Example

Own price 220, competitor avg 190. $\text{Gap} = (220-190)/190 = +0.158$. $\text{C} = 1 + 0.2 \times 0.158 = 1.032$. $\text{e}_{\text{eff}} = -1.5 \times 1.032 = -1.55$. Demand at 220 = $100 \times (220/200)^{-1.55} = 84.6$ units vs 85.7 without competitor adjustment.

F4 — Promotions / marketing (M)

Criticality: MEDIUM — Promotions contaminate raw e . Always flag promo periods before fitting regression.

Dynamic weight formula

$$\text{Q}_{\text{promo}} = \text{Q}_{\text{projection}} \times (1 + \text{lift}_{\text{M}})$$

$$\text{lift}_{\text{M}} = \text{b}_{\text{promo}} \text{ from regression of } \log(\text{Q}) \text{ on } \text{promo_flag}$$

Exclude promo periods from baseline e estimation, OR include promo_flag as covariate. Never mix.

How to find the value

Flag def	is_promo = 1 if: discount > 5% OR campaign live OR sale event active.
Lift range	Email: +5-15%. Paid social: +10-25%. Flash sale: +40-100%.
No promo	Set M = 0. Do not include promo factor in formula — it drops out cleanly.
AI rule	If promo and price change happen simultaneously, model them in separate scenarios.

Data source

- Internal: campaign calendar, promo discount log, marketing spend records.
- Kaggle: Corporacion Favorita Grocery Sales — includes promotion flag column.

Example

Email campaign active. $b_promo = 0.12$ (estimated). $lift_M = e^{0.12} - 1 = 12.75\%$. Base projection at 220 = 85.7 units. Adjusted = $85.7 \times 1.1275 = 96.6$ units. Revenue = $220 \times 96.6 = 21,252$. Promo recovers the demand lost to the price increase.

F5 — Inventory / stock signal (I)

Criticality: LOW-MEDIUM — Stockouts censor demand. Low visible stock creates scarcity uplift. Both distort e .

Dynamic weight formula

$$\text{coverage} = \text{units_in_stock} / \text{avg_daily_sales}$$

$$I = 1.15 \text{ if coverage} < 7 \text{ days (scarcity uplift visible to buyer)}$$

$$I = 1.0 \text{ if } 7 \leq \text{coverage} \leq 30 \text{ (normal)}$$

For e estimation: exclude observations where coverage < 3 days (censored demand — not zero demand).

How to find the value

Coverage	units_in_stock / avg_daily_sales_last_14d. Pull from WMS/ERP daily.
Correction	Periods where stock hit zero — interpolate demand, do not record as zero.
AI rule	If R2 from stock model < 0.03, inventory is not material. Set $I = 1.0$ and exclude.

Data source

- Internal: daily inventory snapshot from WMS/ERP (SAP, NetSuite, Shopify inventory API).
- No direct Kaggle proxy — use any dataset with stock availability columns.

Example

Coverage = 5 days. Low-stock badge visible to buyer. $I = 1.15$. Base projection $85.7 \times 1.15 = 98.6$ units at price 220. Revenue = 21,692. If stock runs out mid-period, apply a supply cap: captured demand = $\min(Q_{\text{new}}, \text{available_stock})$.

F6 — Product lifecycle (L)

Criticality: MEDIUM — Launch and decline phases have materially different e than mature products.

Dynamic weight formula

$$L \text{ modifier: Launch } (<6\text{m}) = 0.7 \mid \text{Growth } (6\text{-}18\text{m}) = 0.85 \mid \text{Mature } (>18\text{m}) = 1.0 \mid \text{Decline} = 1.2$$

$$e_{\text{lifecycle}} = e \times L$$

If product < 3 months old, mark all e estimates PROVISIONAL. Use category proxy e .

How to find the value

Launch date	From product master table in ERP or PIM system.
Phases	Launch <6m. Growth 6-18m. Mature >18m. Decline = sales drop >20% YoY for 2+ quarters.
AI rule	Continuous form: include log(age_months) x log(P) interaction in regression for smoother L.

Data source

- Internal: product master — sku_id, launch_date, category.
- Kaggle: Amazon product datasets with listing date columns.

Example

Product launched 4 months ago -> Launch phase -> $L = 0.70$. $e_eff = -1.5 \times 0.70 = -1.05$. At price 220: $Q = 100 \times (220/200)^{-1.05} = 89.5$ units. Less price-sensitive than a mature product — buyers are still forming preferences.

F7 — Consumer sentiment (X)

Criticality: LOW — Only material during macro shocks. Treat as sensitivity band, not primary driver.

Dynamic weight formula

$$X = 1 + 0.1 \times ((CCI_current - CCI_baseline) / CCI_baseline)$$

$$e_sentiment = e \times X$$

If $R^2 < 0.05$ on sentiment model, set $X = 1.0$ and exclude from weights. This is common.

How to find the value

CCI source	RBI Consumer Confidence Survey (India). OECD Consumer Confidence Index (global). Monthly.
Alternative	Google Trends index for category search volume — proxy for consumer intent.
AI rule	If CCI drops >10 points in one month, floor X at 0.97 regardless of regression.

Data source

- RBI database — Consumer Confidence Survey, monthly CSV.
- OECD.Stat — Consumer Confidence Indicator, all countries.
- Google Trends API (pytrends) — category search volume as proxy.

Example

$CCI_current = 95$, $CCI_baseline = 105$. $X = 1 + 0.1 \times ((95-105)/105) = 0.9905 \sim 1.0$. Negligible. If CCI drops to 80: $X = 1 + 0.1 \times (-0.238) = 0.976$. Apply $e \times 0.976$ — slightly more elastic.

3. Spike detection and weight correction

AI SYSTEM RULE: Run spike detection before every model fit. Spikes that enter regression unchecked corrupt e — the model will underestimate true price sensitivity because demand appeared high even at high prices during the spike. Never fit on raw uncleaned data.

Detection rules

Rule	Condition	Threshold	Flag type
S1	Weekly demand vs rolling 4-week mean	Z-score > 2.0	Statistical spike
S2	Week-on-week demand change	> +80% in 7 days	Velocity spike
S3	Spike with no price drop, promo, or seasonal peak	All 3 explanations absent	Unexplained spike
S4	Google Trends or social index surge	> 3x baseline same week	Hype spike
S5	Demand reverts to pre-spike baseline	Within 3 weeks	Confirmed transient

Classification and treatment

Type	Signal	Action	Weight effect
Type A — Transient hype	Reverts within 3 weeks. Social signal present. No product change.	Exclude from regression. Winsorise or apply decay: $Q_{adj}(t) = Q_{base} + (Q_{peak} - Q_{base}) \times \exp(-0.5 \times t)$	No change to weights — spike is noise.
Type B — Structural shift	New baseline 2x above pre-spike, holds for 4+ weeks. New buyer cohort visible.	Re-baseline model. Discard all pre-break data. Re-estimate all factors.	Reset all w_i from new data. Weights will shift significantly.

DEFAULT RULE: If spike type is ambiguous, classify as Type A (penalise). It is safer to exclude a genuine structural shift than to corrupt e with a transient spike. The structural shift will become apparent in the next 2-3 data batches.

4. Factor reliability assessment

Not all modelled factors are equally trustworthy. The table below shows estimated reliability for each factor and the primary reason it can fail. Low-reliability factors remain in the model because they matter in edge cases — but the dynamic R2-weighting system will automatically downweight them when their explanatory power is low.

Factor	Reliability	Primary failure mode	AI system behaviour when unreliable
F1 Price elasticity	88% HIGH	Thin data (<30 obs) or promo noise corrupts e	Flag as PROVISIONAL. Use category proxy e.
F2 Seasonality	82% HIGH	Unreliable on new products with <2 seasonal cycles	Set S = 1.0 if fewer than 2 full cycles available.
F3 Promotions	65% MEDIUM	Brand vs promo effect mixed — attribution is hard	Include promo_flag as covariate; report lift with CI.
F4 Inventory signal	60% MEDIUM	Scarcity uplift is behavioural — hard to isolate in data	Only apply if badge is confirmed visible to buyer.
F5 Product lifecycle	58% MEDIUM	Phase boundaries are heuristic; vary widely by category	Use continuous log(age) x log(P) interaction instead.
F6 Competitor pricing	52% MEDIUM	Scraped prices lag reality; bundles and offers not captured	If data >3 days old, set C = 1.0 (neutral).
F7 Consumer sentiment	38% LOW	CCI is national — rarely explains individual SKU demand	Set X = 1.0 unless CCI move >10 pts in one month.

RELIABILITY SCORE = estimated average R2 contribution to demand variance across product categories. Scores are indicative. Run SHAP value analysis on your specific product category to validate. A factor with reliability 38% still belongs in the model — its weight will be low, but when it does fire (macro shock, viral moment) it matters enormously.

5. Human-judged and unquantifiable factors

The following factors affect demand materially but cannot be reliably included in an automated formula. They are split into two tiers. H-tier factors can be partially measured but their relationship to demand is too unstable to regress on. U-tier factors have no consistent data signal at all. Both tiers require a human decision before a pricing action is approved.

H1 — Brand perception

Tier: HUMAN-JUDGED — Proxies exist but do not reliably predict demand response to price changes.

Why it cannot be modelled automatically

Brand equity affects willingness to pay, but NPS and review scores are lagging indicators. A single viral PR event can collapse or spike demand within hours — far faster than any weekly model can detect. The relationship between NPS and price elasticity is product-specific and unstable.

What data exists (partial proxies only)

Proxy metrics	NPS score, star rating average, review count velocity, social sentiment score (VADER/BERT).
Stability	Useful as a flag but not as a regression covariate — too noisy and lag-heavy.
AI limitation	Model should PAUSE and require human review if NPS drops >10 points in 2 weeks.

Where to find it

- Internal: customer survey data, NPS tracking tool (Medallia, Qualtrics).
- External: review scraping from Amazon/Google/Trustpilot, brand mention monitoring (Brandwatch, Mention).

Human checklist question

Is there a recent PR event, viral complaint, or reputational shift that the model cannot see?

Signal to watch

Watch: NPS trend (weekly), star rating delta, press mention volume, social sentiment score.

Example

Product has NPS of 62. A manufacturing defect story appears in a major publication. NPS drops to 41 over 3 weeks. The demand model would project normal elasticity from price changes. In reality, demand is suppressed regardless of price — customers are avoiding the brand entirely. Human must override model and hold price flat until brand recovers.

H2 — Substitute availability

Tier: HUMAN-JUDGED — Cross-price elasticity requires competitor sales data that is rarely accessible.

Why it cannot be modelled automatically

How easily buyers can switch to another product is the most important context for own-price elasticity. But estimating cross-price elasticity requires competitor unit sales volume — data that is almost never available. Substitute availability also changes discretely (a new competitor launches on a specific date) rather than gradually, making regression-based detection slow and unreliable.

What data exists (partial proxies only)

Proxy	Number of competing SKUs in category, search share data (SEMrush), share of shelf.
Change signal	New entrant launch date — a step-change in substitutability, not a smooth curve.
AI limitation	Model should flag if category SKU count increases >20% vs prior quarter.

Where to find it

- Internal: category assortment data, search keyword share reports.
- External: Nielsen/Kantar category reports, Google Shopping category listings, retail audit data.

Human checklist question

Has a strong new competitor or meaningfully cheaper substitute entered the category recently?

Signal to watch

Watch: new entrant product listings, category market share reports (monthly), SEMrush search share delta.

Example

A well-funded D2C competitor launches a nearly identical product at 15% lower price. Cross-price elasticity of own product is now approximately -2.5 (highly elastic to competitor price). The demand model only knows own price history — it will significantly overestimate retention at a price increase. Human must manually increase e_{eff} estimate or delay the price increase.

H3 — Channel and placement

Tier: HUMAN-JUDGED — Confounds price experiments. Cannot separate placement effect from price effect.

Why it cannot be modelled automatically

Where a product appears — organic search rank, shelf position, algorithm-recommended slot — can shift demand by 30-60% independently of price. If a product moves from page 1 to page 3 of search results at the same time as a price change, the demand model will attribute the drop to price when placement caused it. You cannot A/B test price and placement simultaneously and isolate either effect cleanly.

What data exists (partial proxies only)

Proxy	Organic search rank (SEMrush, Ahrefs), listing position, ad impression share.
Timing	Track any rank or placement change in the 2 weeks before and after a price change.
AI limitation	Model should block price change execution if organic rank has moved >5 positions in 14 days.

Where to find it

- Internal: search rank tracking tool, retailer dashboard, listing audit log.
- External: SEMrush, Ahrefs, Helium10 (Amazon), Jungle Scout.

Human checklist question

Will a search rank change, de-listing, or placement shift coincide with or follow this price change?

Signal to watch

Watch: organic rank delta (weekly), ad impression share, retailer in-stock and featured placement status.

Example

A price increase of 8% is implemented. At the same time, an algorithm change drops the product from position 3 to position 9 in category search. Demand falls 35%. The demand model attributes this to price elasticity and estimates $e = -4.4$. True e is approximately -1.5 — the placement change caused most of the drop. Future projections using the corrupted e will be badly wrong.

H4 — Regulatory and policy risk

Tier: HUMAN-JUDGED — Binary cliff-edge events. Model cannot predict regulatory timing or scope.

Why it cannot be modelled automatically

Price controls, mandated discounts, import tariffs, and sector-specific regulations can instantly invalidate a pricing strategy. These events are binary (the regulation either applies or it does not) and the timing is determined by political processes, not data. No regression model can anticipate them. In regulated sectors (pharma, energy, telecoms, food staples), this factor is often the binding constraint.

What data exists (partial proxies only)

Monitoring	Government gazette, sector regulator publications, trade association alerts.
Lead time	Draft regulations typically have 30-90 day comment periods — catchable with monitoring.
AI limitation	Model must have a hard price ceiling field per SKU that overrides any projection output.

Where to find it

- Internal: legal/compliance team regulatory calendar, government relations tracker.
- External: Government gazette (India: egazette.gov.in), sector regulator websites, trade association newsletters.

Human checklist question

Is there pending legislation, a price ceiling, tariff, or sector regulation that would cap or override this projection?

Signal to watch

Watch: government gazette, regulator press releases, trade association alerts, parliamentary committee reports.

Example

A food staples category is facing proposed price ceiling legislation. The demand model projects that a 12% price increase maximises profit. The proposed ceiling would cap the price 6% below the current price. Implementing the model's recommendation would immediately violate the regulation once it passes. Human must hold all price increases in this category until regulatory outcome is confirmed.

U1 — Perceived fairness of price

Tier: UNQUANTIFIABLE — Psychological. No data signal until backlash arrives. Never model this.

Why it cannot be modelled automatically

Whether buyers perceive a price as 'fair' or 'exploitative' is determined by context, framing, and social norms — not by the price itself. The same price increase that is accepted in normal conditions can trigger viral backlash during a shortage, crisis, or public emergency (surge pricing controversy). The backlash is sudden and non-linear: demand can drop 60% in 48 hours from social pressure alone. There is no data signal to detect this before it happens.

What data exists (partial proxies only)

Proxy	Social sentiment analysis — but this only measures backlash after it starts, not before.
Context flag	Check: is there an active public crisis, shortage, or emergency in the category?
AI rule	NEVER implement a price increase during a declared emergency or widely-covered shortage. Hard block.

Where to find it

- Social listening tools: Brandwatch, Sprinklr, Mention — real-time sentiment.
- News monitoring: Google Alerts, media monitoring service for category keywords.
- No quantitative data source predicts this reliably before the fact.

Human checklist question

Does this proposed price feel exploitative given the current news context — shortage, crisis, or emergency?

Signal to watch

Watch: social sentiment score, news tone for category, active public health or supply emergency declarations.

Example

Essential goods category. A regional supply disruption causes stock tightness. The demand model correctly identifies inelastic demand ($e = -0.4$) and projects that a 25% price increase would increase profit by 40%. In reality, implementing this during a supply crisis triggers news coverage about price gouging. Demand collapses 70% as buyers actively boycott. Human must veto the model's recommendation on ethical and reputational grounds.

U2 — Social contagion and word of mouth

Tier: UNQUANTIFIABLE — Causality is inverted. Demand drives virality, not the other way around.

Why it cannot be modelled automatically

Viral demand spread through social networks cannot be included in a price-demand model because the causal direction is inverted: virality causes demand, not price. Including social volume as a predictor of demand in a price regression would confuse correlation with causation and produce a model that thinks posting on TikTok is a substitute for pricing. Additionally, virality is inherently unpredictable — the same product features can go viral or not depending on which influencer happens to post that week.

What data exists (partial proxies only)

Proxy	Google Trends index, TikTok/Instagram mention count, hashtag velocity.
Use	These proxies are useful for spike DETECTION (Section 3) but not for demand PROJECTION.
AI rule	If social volume is >3x baseline, trigger spike detection review. Do not feed into formula.

Where to find it

- Google Trends API (pytrends) — free, weekly/daily granularity.
- Social listening: Brandwatch, Sprinklr, TikTok Creator Marketplace data.
- No source reliably predicts viral moments before they occur.

Human checklist question

Is there an active viral moment driving demand that will collapse without warning when attention shifts?

Signal to watch

Watch: Google Trends velocity (week-on-week change), TikTok/Instagram mention count, influencer post volume.

Example

A skincare product is organically featured in a viral TikTok video. Demand spikes 4x in 3 days. The spike detection system correctly flags this as a potential Type A (transient) or Type B (structural) spike. A human analyst reviews: if a new customer cohort is visible in acquisition data, reclassify as Type B and re-baseline. If demand returns to prior level within 3 weeks, classify as Type A and exclude from model. At no point should the formula treat the viral moment as a predictable input.

6. Human judgment checklist

Run this checklist before every pricing decision. If ANY question is answered YES or UNKNOWN, a human analyst must review before the price change is implemented. The model output is advisory only until all six questions are cleared.

Factor	Question to answer before pricing	If YES or UNKNOWN	Signal to check
H1 Brand perception	Is there a recent PR event, viral complaint, or reputational shift?	Hold price change. Investigate.	NPS trend, star rating delta, press mentions
H2 Substitute availability	Has a strong new competitor or cheaper substitute entered recently?	Manually increase e_eff estimate.	Category SKU count, search share delta, new entrant launch date
H3 Channel and placement	Will a rank or placement change coincide with this price change?	Separate experiments. Do not proceed simultaneously.	Organic rank delta, ad impression share, retailer placement
H4 Regulatory risk	Is there pending legislation or a price ceiling in this category?	Block price change. Escalate to legal.	Government gazette, regulator website, trade association alerts
U1 Perceived fairness	Could this price increase be perceived as exploitative given current events?	Veto. Reputational risk outweighs profit gain.	Social sentiment, active shortage/crisis news, category context
U2 Word of mouth	Is active virality driving current demand that will collapse soon?	Run spike detection. Classify Type A or B before projecting.	Google Trends velocity, TikTok/Instagram mention count

AI SYSTEM RULE: The AI pricing system must surface this checklist to a human approver before executing any price change greater than 5%. Responses to all 6 questions must be recorded and logged with the pricing decision. A fully automated price change with no human sign-off is not permitted under this framework.

7. AI system integration notes

This section is written specifically for engineers and AI systems implementing this framework. It defines the execution order, fallback rules, and hard stops that must be built into any automated pricing pipeline.

Execution order

Step	Action	Input	Output / condition to proceed
1	Spike detection	Raw weekly demand series	Clean series or classified spikes (Type A / B)
2	Data cleaning	Spike-classified series	Training data with Type A periods excluded
3	Estimate e via rolling OLS	Clean log(P), log(Q) pairs	e with 95% CI. Flag if e > 5 or CI width > 1.5
4	Compute factor values	External data per factor	S, C, M, I, L, X values (default 1.0 if missing)
5	Compute R2 per factor	Regression residuals + factor data	w_i weights, normalised to sum = 1.0
6	Project Q_new	P_new, all factor values, e_eff	Q_new with P10/P50/P90 scenario band
7	Compute profit curve	Q_new across price range, Cost	Optimal price point and profit at P_new
8	Surface human checklist	H1-H4 and U1-U2 inputs	Human sign-off required before proceeding
9	Execute price change	Approved price	Log decision with all factor values and weights

Hard stops — AI must not proceed without human review

e > 5.0	Elasticity estimate is likely corrupted. Re-clean data and re-estimate.
CI width > 1.5	Insufficient data. Use category proxy e and mark output PROVISIONAL.
Uncategorised spike	A flagged spike that has not been classified Type A or B. Block model run.
Price change > 25%	Outside safe extrapolation range of most e estimates. Require manual review.
CCI drop > 10 pts/month	Macro shock. Apply X floor and flag all projections as highly uncertain.
Any H/U checklist = YES	Human must review and sign off. Automated execution blocked.

VERSION AND AUDIT LOG: Every model run must record: timestamp, sku_id, all factor values used, all R2 weights, e estimate and CI, P_new, Q_new projection, profit projection, spike classifications, and human approver name. This log is required for model governance, regulatory audit, and future re-training.